AquariusCodaExp: A 48M-Parameter One-Step Pixel-Space Denoiser

JunyiDu2009
AquariusDiffusion / AquariusImage · October 2026

Abstract Diffusion models require multi-step iterative sampling, so inference latency scales with step count. We train a 48.1M-parameter pixel-space UNet that performs **single-step direct-regression** denoising on images corrupted with noise of variance ratio p∈[1%,50%] — no VAE, no sampling chain, no teacher distillation. We align the training and inference domains with multi-resolution × aspect-ratio buckets (11 buckets incl. 16:9), unlock noise levels via a four-stage curriculum, and converge with a plateau early-stop gate at 76K steps (saving ≈40 GPU-hours against a fixed 250K-step budget). After training, weights are exported to NVFP4 (E2M1 + per-16 E4M3 scale), shrinking 96 MB to 27.2 MB (3.5×). On COCO val2017, the NVFP4 dequantized weights reach 28.54 / 25.78 / 24.07 dB PSNR and 0.191 / 0.270 / 0.326 LPIPS at p=10/30/50%. All training and inference ran on a single RTX 5060 8GB.

1 Introduction

Diffusion models lead in generation and restoration quality, but their inference cost scales linearly with sampling steps, which is prohibitive on low-power devices. A pragmatic alternative for bounded, known corruption is a feed-forward denoiser: when the noise strength is known or estimable, restoration can be completed in **one forward pass**. We study exactly this setting — inputs are "noisy but roughly clean" images (variance ratio p∈[1%,50%]), and a single-step model must recover the clean image.

Our contributions: (1) an open implementation of a 48.1M pixel-space UNet that trains end-to-end on an 8GB consumer GPU; (2) a concrete recipe of multi-resolution × aspect-ratio bucketing and a four-stage noise curriculum; (3) a hyperparameter-free **plateau early-stop** gate that fired at exactly 76K steps in practice; (4) a post-training NVFP4 export pipeline verified live in an interactive demo.

2 Relation to Prior Work

Consistency-model and distillation approaches compress multi-step sampling into one or two steps, but usually target generation and require a teacher. We instead treat denoising as regression conditioned on p, so the training objective maps one-to-one onto the deployed kernel. Unlike AquariusTerimageExp in this repository — whose weights live in quantized space from step 0 — this experiment deliberately uses FP16 training with a post-training NVFP4 export, serving as a conservative baseline for quantization gains.

3 Method

3.1 Task

Noising follows $x_p = \sqrt{(1-p)}\cdot x_0 + \sqrt{p}\cdot\varepsilon$ with p the noise variance ratio. The model takes (x_p, p) and predicts x_0 in a single step.

3.2 Network and Training

The 48.1M UNet uses GroupNorm + SiLU, mid-block multi-head self-attention, sinusoidal injection of p, and gradient checkpointing. Loss is MSE + 0.1·LPIPS(VGG) on COCO 2017 (115K images).

Buckets. Training samples one of 11 resolution×aspect buckets (256²–640×384, incl. 16:9/9:16) per step with pixel-budget-scaled batches, so the inference domain covers any /16-divisible size. **Curriculum.** Four stages progressively unlock 1–10% → +20% → +30% → 1–50%. **Early-stop.** The gate averages LPIPS at p=10/30/50%; after 20 consecutive evaluations (10K steps) improving ≤0.002, the run saves and exits. It fired at 76,000 steps, saving ≈40 GPU-hours.

3.3 NVFP4 Quantization

Post-training export to NVFP4: E2M1 nibble weights + one E4M3 block scale per 16 channels + an FP32 tensor scale. Inference uses the dequantized path. Size drops 96 MB → 27.2 MB (3.5×) with metrics essentially unchanged (Table 1).

4 Experiments

4.1 Setup

Validation on COCO val2017 (5K; 128 images for fast eval). Everything ran on a single RTX 5060 8GB (sm_120) with a torch memory cap (--mem-frac 0.65), allocation thresholds, and periodic cache flushes to avoid Windows WDDM shared-memory paging.

Noise	PSNR (dB) ↑	LPIPS ↓
p=1%	33.50	0.091
p=10%	28.54	0.191
p=20%	26.86	0.236
p=30%	25.78	0.270
p=40%	24.90	0.297
p=50%	24.07	0.326

Table 1. Final metrics on the full val2017 (step 76,000, NVFP4 dequantized).

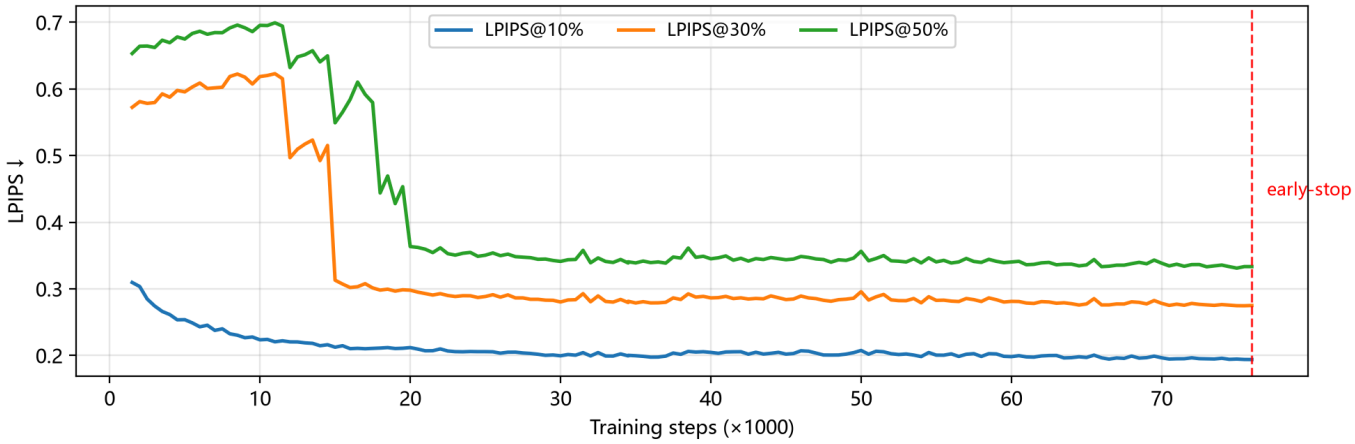


Figure 1. Validation LPIPS convergence. All three tiers plateau after 60K steps; the gate triggers early-stop at 76K (red dashed).

4.2 Qualitative Results: One Image, Five Noise Levels

Figures 2–4 show full comparisons on single validation images: top rows are noised inputs, bottom rows are model outputs, each cell annotated with measured PSNR and LPIPS. At p=10% outputs are near-lossless; at p=30% texture softens while structure holds; at the p=50% extreme the global structure survives with clearly softened high frequencies — the capacity limit of a 48M single-step regressor, suggesting larger models or few-step refinement for higher fidelity.

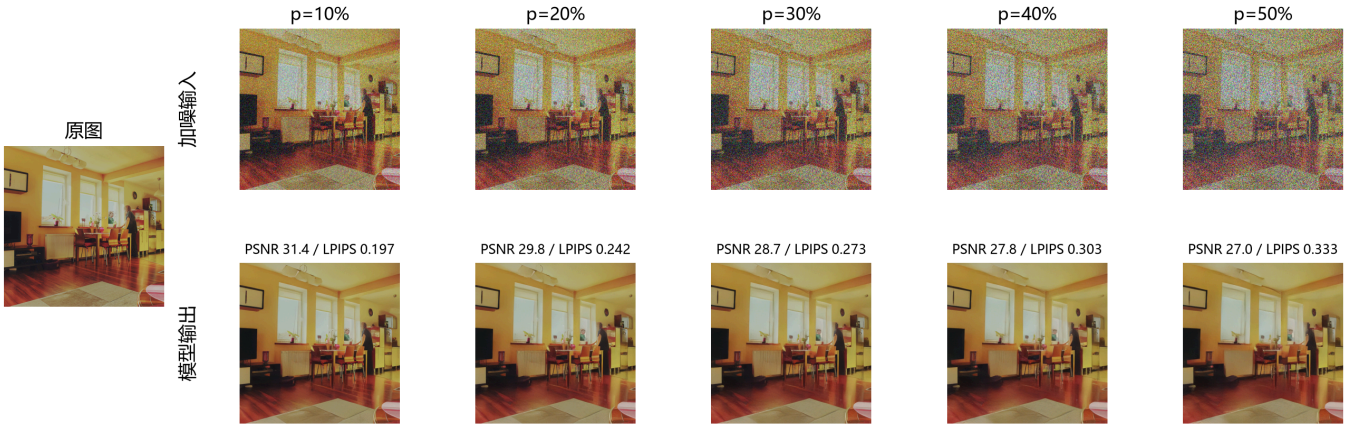


Figure 2. Sample 1: noised inputs (top) vs. model outputs (bottom) across five noise levels.

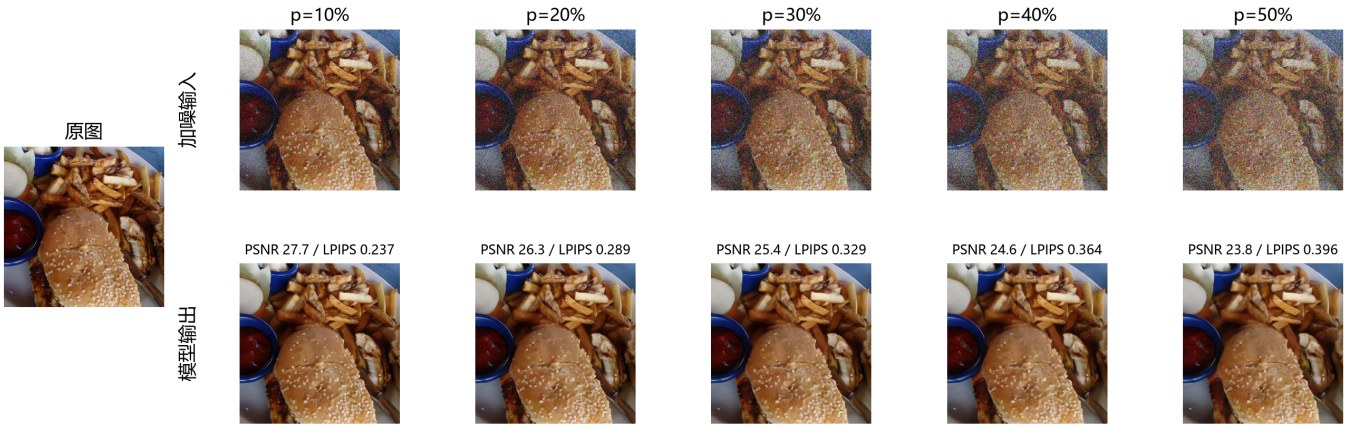


Figure 3. Sample 2: even at p=50%, the disc structure is recovered in one step.

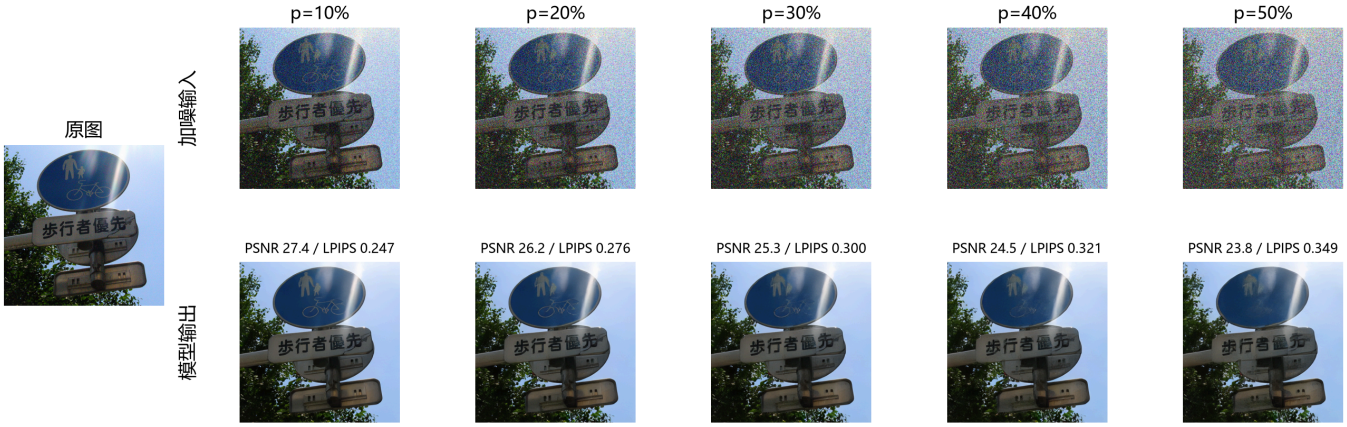


Figure 4. Sample 3: texture softens at extreme noise, while exposure and color cast are corrected.

5 Conclusion and Limitations

Within the bounded noise regime p∈[1%,50%], a 48M single-step regressor delivers usable denoising quality with near-lossless NVFP4 quantization. Limitations: the absolute LPIPS remains behind perceptual acceptance targets; extreme tiers soften texture; and native low-bit training was not explored here. Follow-ups include recalibrated acceptance lines and FP8/FP4 QAT comparisons.

References

1. Zhang et al. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. CVPR 2018.
2. Lin et al. Microsoft COCO: Common Objects in Context. ECCV 2014.
3. Ho et al. Denoising Diffusion Probabilistic Models. NeurIPS 2020.
4. Song et al. Consistency Models. ICML 2023.
5. NVIDIA. NVFP4 Format Specification. Open Compute Project, 2025.
6. JunyiDu2009. AquariusTerimageExp: Native Ternary Text-to-Image. 2026. In this repository, AquariusImage/.